

Prototyping

(summary slides)

Prototyping

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Introduction to prototyping

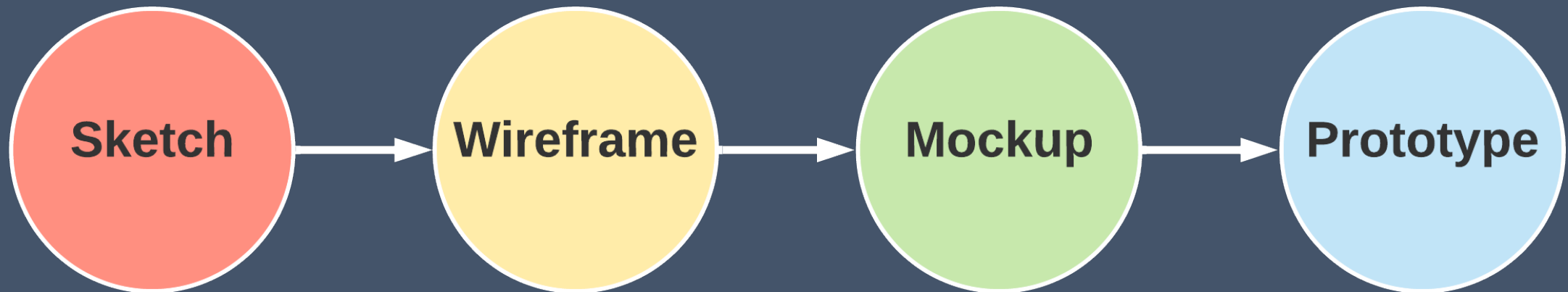
Sketches, wireframes, mockups, and prototypes

Lecture: Sketches, wireframes, mockups, and prototypes

Module: Introduction to prototyping

Introduction

- **Product designers** often hear the terms **sketch**, **wireframe**, **mockup**, and **prototype**.
 - ✓ Although the terms are related, they are different.
- In this **lecture**, we want to **review** them.



Sketch

Sketch

- A **sketch** is a **freehand drawing** on a **piece of paper** or in a **digital tool** that gives you a **basic representation** of your **concept**.
- **Sketches** can be very **helpful** during the **conceptualization phase** of your **design process**.

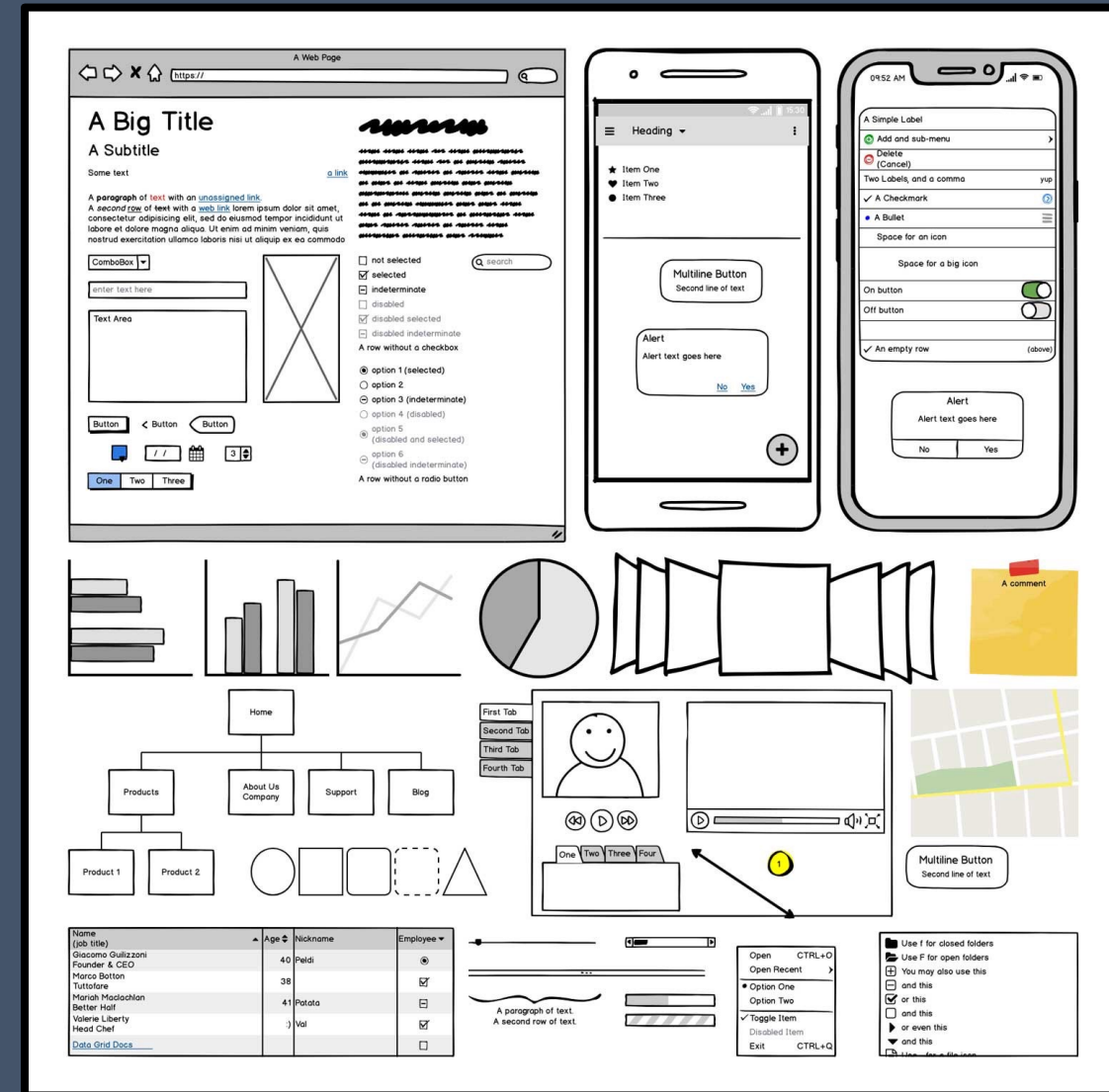


Source: <https://bit.ly/39AfIM5>

Wireframe

Wireframe

- low-fidelity design artifacts that represents the layout of the content in a user interface.
 - ✓ Usually in grayscale or black and white.
- Can be created using pen & paper or with some tool like Balsamiq.
- Useful in an early stage of the product/software design process.



Source: <https://balsamiq.com/wireframes/>

Mockup

Mockup

- **Medium or high-fidelity visualization** of your design.
- A mockup delivers the **visual look** of the **product design**, and they are **great** for **evaluating the look and feel** of your design.
- A mockup looks like the **final product**.



Source: <https://bit.ly/39JbuSe>

Prototype

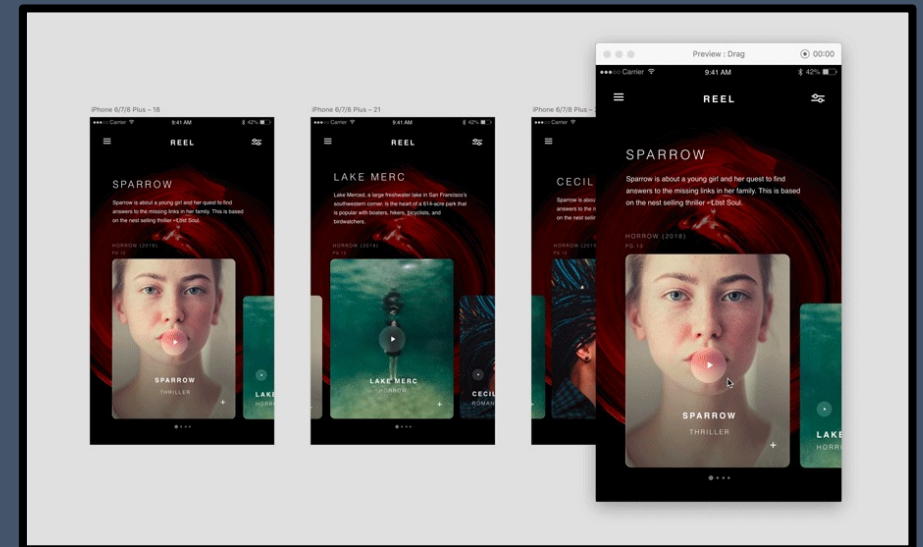
Prototype

- **Early working model** of a **design** used to gain **feedback** and **quickly experiment** with **new ideas**.
- A **prototype** is done with a **mix** of **sketches**, **wireframes** and **mockups**.
 - ✓ However, **unlike sketches, wireframes and mockups, prototypes are always interactive.**



Why do designers need to prototype?

- **Simulate the interaction between the user and the interface.**
 - ✓ Making sure that the design flows and user interface make sense to the users.
- **Validate the design(ideas and concepts) early.**



Source: <https://adobe.ly/2XQVfxb>

Are prototypes created with code?

- A **prototype** does not have to be made with code.
 - ✓ You can **prototype** with anything from sketches to using fully rendered mockups.
- We will **look at some tools** in this course that will help you create **prototypes** without coding.

HTML prototypes

HTML prototypes

- While some people might feel more comfortable using HTML interactive prototypes, it isn't a recommended approach.
- The goal of prototyping is to get through multiple ideas quickly, and HTML based prototypes can be time consuming to update.

Why do we prototype?

Lecture: Why do we prototype?
Module: Introduction to prototyping

Why do we prototype?

- 1 To limit the expense of development costs
- 2 To test out design concepts
- 3 To refine the usability of the product
- 4 To communicate our designs with members of the project team

Time cost of creating prototypes

Lecture: Time cost of creating prototypes
Module: Introduction to Prototyping

Time cost of creating prototypes

- **Make sure** that you've given your **prototype** some **thought before putting it in front of users**.
 - ✓ Because **prototyping** has a **time cost**.
- The **cost** is **worth** the **investment**.
 - ✓ Because you'll **find** out through the **prototyping process** where your **design ideas fail** and **where they shine**.

Time cost of creating prototypes

- As a **result** of the **cost on time**, when **producing prototypes**, the **idea** is to **get through prototyping the design as quickly as possible**.
 - ✓ To get **better ideas** and **better designs** through feedback.

Prototyping fidelity

The 3 types of prototyping fidelity

Lecture: The 3 types of prototyping fidelity
Module: Prototyping fidelity

Fidelity

- **Fidelity** refers to the **degree of completeness** of the **user interface, interactivity** and **flows**.
- **Fidelity** can refer to the **visual appearance**, but also to the **degree of functionality** represented in a **prototype**.

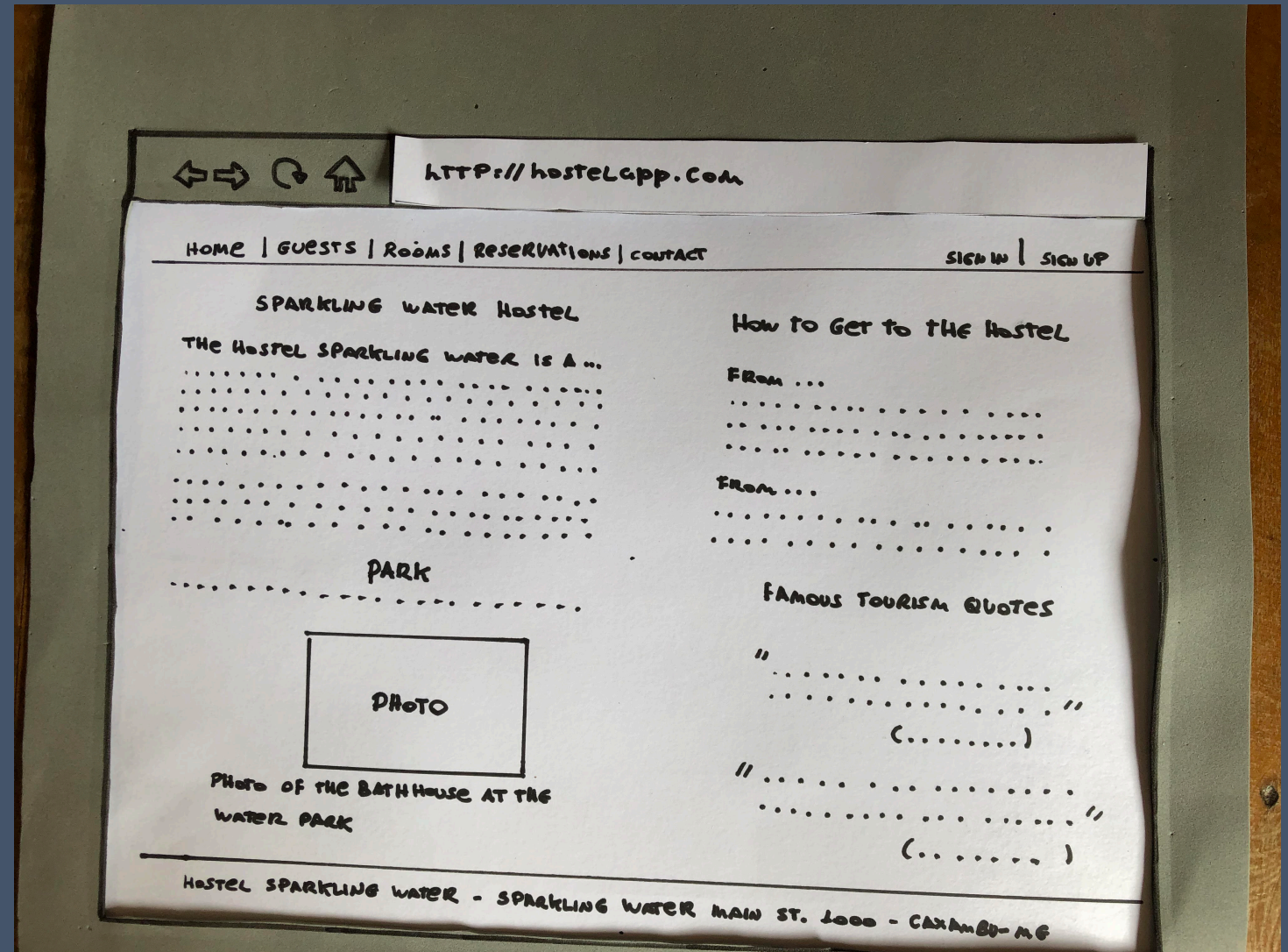
The 3 types of prototyping fidelity

1. **Low-fidelity** prototyping.
2. **Medium-fidelity** prototyping.
3. **High-fidelity** prototyping.

Low-fidelity prototyping

Lecture: Low-fidelity prototyping
Module: Prototyping fidelity

Low-fidelity prototype (sketch)



Homepage of the hostel Sparkling Water

Low-fidelity prototypes

- Sometimes **defined as only paper-based**.
- In this **course** we define **low-fidelity** to include **clickable wireframes** that can be **created** in a **prototyping software**.
 - ✓ With **minimal interactions** like **linking screens together**.

Advantages of low-fidelity prototypes

Advantages of low-fidelity prototypes

- Easy to create with little time or effort.
- You can **update** your **sketches** and **create new sketches** to get more **feedback** and **rapidly iterate** through the **concepts** and **incorporate** the **users' feedback**.
- **Users are more likely** to be **open** with their **feedback**.
 - ✓ If you **ask for feedback**, **users** will be **more likely** to **give** their **honest opinion** because it's **obvious** that **design** is in an **early state**.

Disadvantages of low-fidelity prototypes

Disadvantages of low-fidelity prototypes

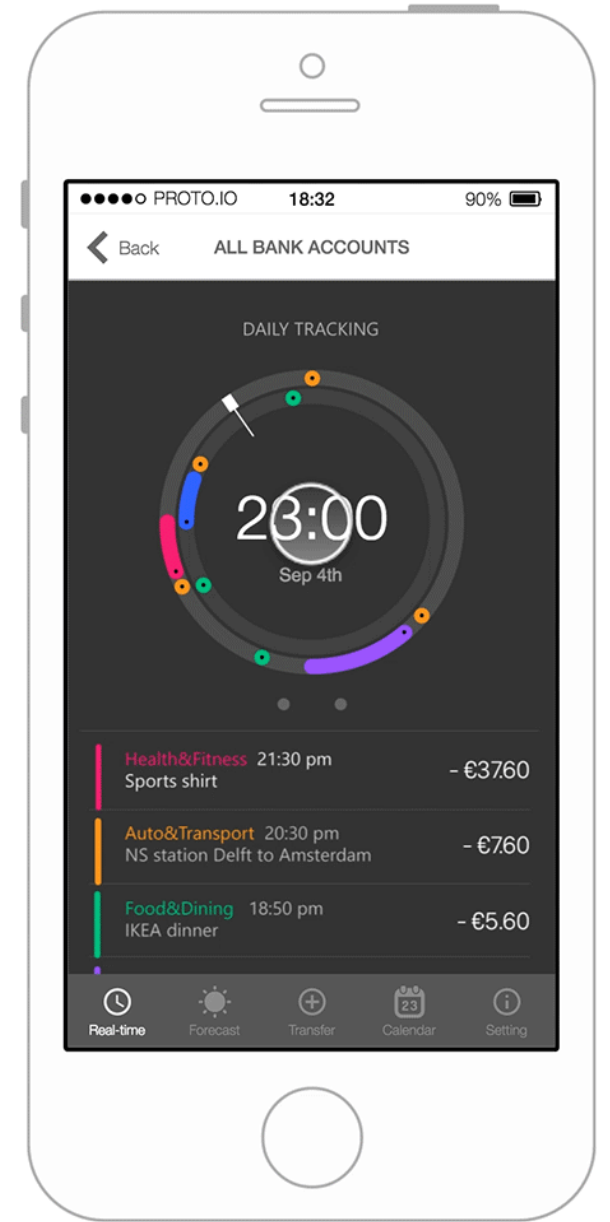
- **There's very limited flows and interaction behavior represented in the design.**
- **The design of low-fidelity prototypes limits realistic interactions.**
 - ✓ You won't receive rich feedback on how the interface elements work.
 - ✓ You won't receive rich feedback on how all the flows will tie together.
- **A facilitator is required to walk the user through the design.**

Medium-fidelity prototyping

Lecture: Medium-fidelity prototyping
Module: Prototyping fidelity

Medium-fidelity prototype

Medium-fidelity prototype



Source: <https://bit.ly/3nOE1Lb>

Advantages of medium-fidelity prototypes

Advantages of medium-fidelity prototypes

- **Medium-fidelity prototypes are more representative of the look and feel of the design.**
 - ✓ However, they may not have all the **functionality** and **flows** ready.
- Because the **design feels more real**, users will go through the **prototype** as they would a **normal website/application**.

Disadvantages of medium-fidelity prototypes

Disadvantages of medium-fidelity prototyping

- Need to **facilitate** some of the **design** because you'll **want feedback** on the **last parts** of the **designs** that are **unfinished**.
- **Medium-fidelity prototypes** take **more time** to **create** and they're also very **difficult** to **update**.

High-fidelity prototyping

Lecture: High-fidelity prototyping
Module: Prototyping fidelity

High-fidelity prototype



Source: <https://bit.ly/3iniYyq>

Advantages of high-fidelity prototypes

Advantages of high-fidelity prototypes

- As a result of the finished appearance of the design, most usability issues at this level will be caught.
 - ✓ You'll get feedback on everything including content, field labels, color choices, and general attractiveness and usability of the product.
- ✓ The product design doesn't need a facilitator for high-fidelity prototypes.

Disadvantages of high-fidelity prototypes

Disadvantages of high-fidelity prototypes

- **High-fidelity prototypes** take **more time** to **create** and they're also **very difficult** to **update**.
- The **users aren't** going to be as **willing** to **give feedback** because the **design looks** so **final**.

Testing prototypes

Prototyping testing sessions

Lecture: Prototyping testing sessions
Module: Testing prototypes

Prototyping testing sessions

- The goal of testing your prototype is to find out whether the designs attend users' expectative and solve their problems.
- It is important to build a prototype to meet the needs of what you're going to test.
 - ✓ Establish a clear goal of what you want out of your test.
 - ✓ build it in a way that won't distract from the outcome you're looking for.

Set user expectation

- **Don't** forget to **set** some **expectations** with **users** on the **kind** of **feedback** you're **looking for**.
 - ✓ If the **prototype** isn't fully functional let them **know** that **some things** just **might not work** yet.

Identify the target audience for testing

- Make sure you're **getting feedback** from your **target audience**.
 - ✓ Need to consider who you'll be testing your **prototype** with.



Source: <https://bit.ly/3nOE1Lb>

Prototyping testing sessions

- A **design process for prototypes** usually has an **iterative lifecycle**.
 - ✓ The **product design evolves** through sessions of **prototyping testing**.



Facilitating prototyping testing sessions

Lecture: Facilitating prototyping testing sessions

Module: Testing prototypes

**The need for a facilitator during
prototyping testing sessions**

Facilitating a testing prototype session

- No matter **how solid** you **think** your **design** is, it's always **important** to **listen** to a **user walk** through it.
 - ✓ This is **accomplished** with the **aid** of a **facilitator**.
- It's **important** the **facilitator** is **skilled** in **asking** the **right questions**.

Examples of questions used during prototyping testing sessions

Lecture: Examples of questions used during prototyping testing sessions

Module: Testing prototypes

Types of questions

Questions used during prototyping testing sessions

Question example (low-fidelity prototypes)

- To get feedback at the concept level using a low fidelity prototype, you can show the user the sketches, and some high-level questions.
- Example:

“What do you think of this idea?”

Question example (medium-fidelity prototypes)

- For a **medium fidelity prototype**, you might show the user a **prototype** that has **some functional links and flows**.
 - ✓ Try to **get some feedback** around **non-functional areas** as well.
- **Example:**

“What do you think this menu item does?”

Question example (high-fidelity prototypes)

- For a high-fidelity prototype, you might ask more specific questions on interactive elements and task flows.
- Example:

“Can you find a room you like and create a reservation for it?”

Design process & models

Overview of user-centered design processes

Lecture: Overview of user-centered design processes

Module: Design process & models

User-centered design process

- A user-centered design process to product development tries to ensure that the products you're developing will delight the user and meet the needs of the business.
- Prototyping is a big part of the user-centered design process.

Prototype goals change with the design phase

- The goal of your **prototype** will **depend** on which **phase** the **design process** you're in.
- **Ideation phase**
 - ✓ **Ideation** is the formation of **ideas** or **concepts**.
 - ✓ **Iterate** through **various ideas** and **concepts**.
 - ✓ **Start** with **paper prototyping**.
- **Delivery phase**
 - ✓ You might have a **more robust functional prototype** for **usability feedback**.

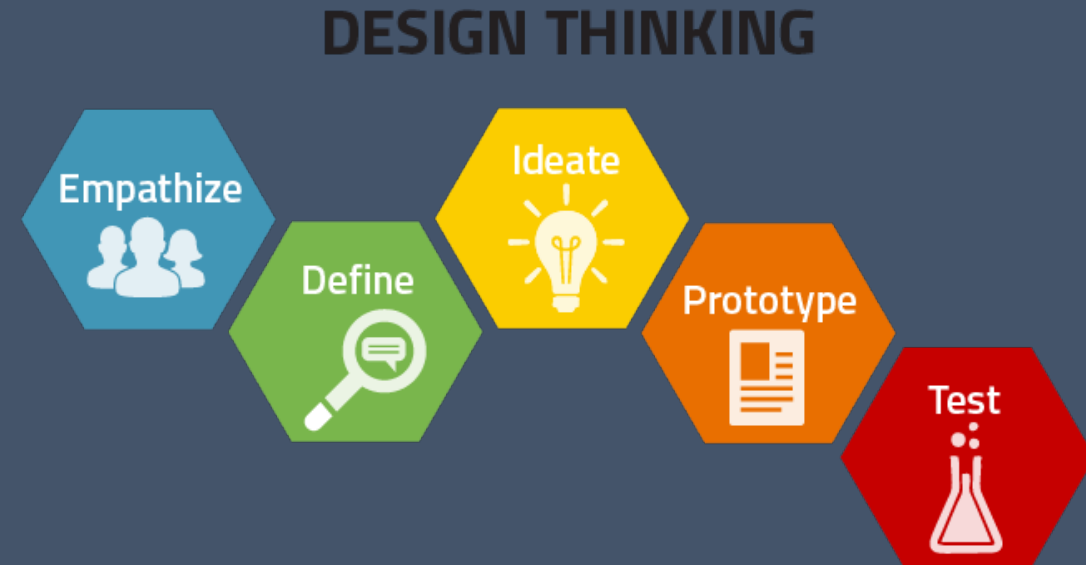
Problem statement

- As you get further in the **user-centered design process**, you'll want to **define the problem statement** to **make sure** that you're **ideating** and **designing** for the **right problem**.
- Once that **problem statement** is **defined**, you can **start thinking** about **design approaches** that **might solve** the **issue**.

Design thinking & Design models

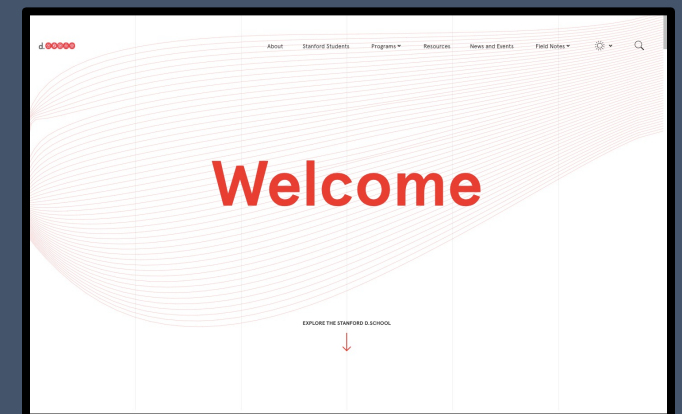
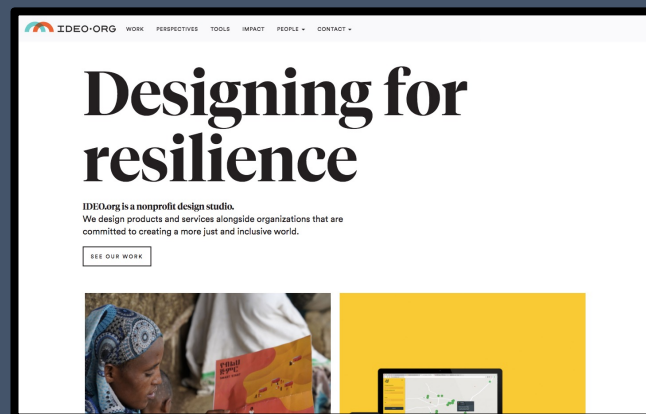
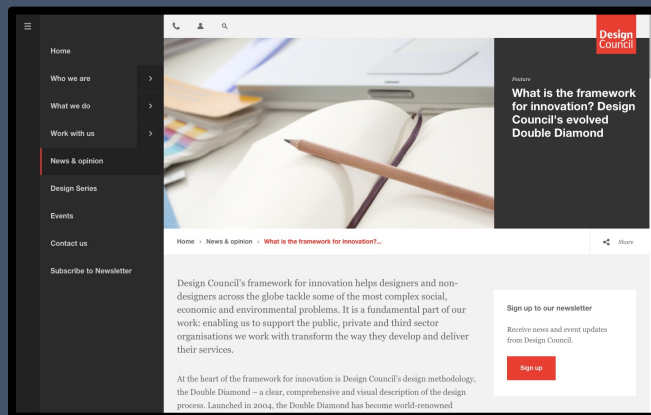
Design thinking & design models

- There are **different models** of the **design process** and **how** to apply **design thinking** to your **product** or to **anything**.
 - ✓ The **key steps** in all **models** are **practically** the same.
- **Design thinking** is a **problem-solving method**.
 - ✓ It is a **way** of **thinking** and **working** as well as a **collection** of **hands-on methods**.



The 3 key design models

1. Design council's double diamond
2. IDEO.org's human-center design approach
3. Stanford d.school's approach



Key attributes of the 3 design models

Key attributes of the 3 design models

- User-centered with their focus on understanding their challenges while discovering opportunities and insights.
- Defining the problem space based on the insights gathered from the users.
- Ideating through various designs concepts and ideas to come up with potential solutions.
- Prototyping and iterating on those design ideas and concepts
- Delivering on the key idea.



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At the heart of the framework for the Double Diamond – a clear, collaborative process. Launched in 2004, the Double Diamond has become world-renowned

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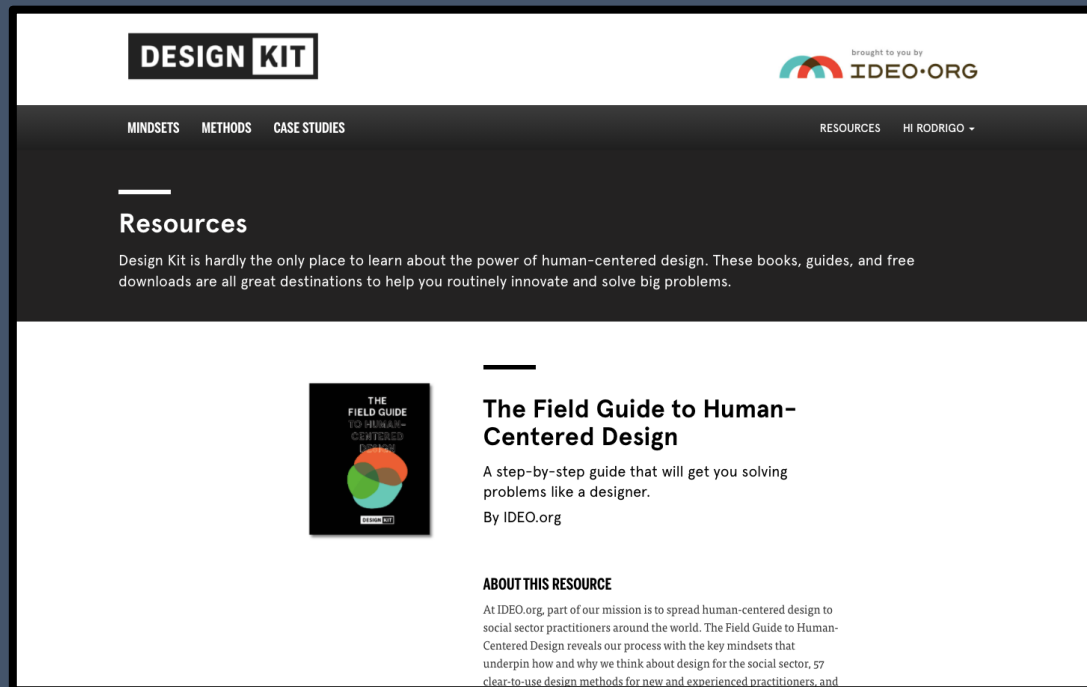
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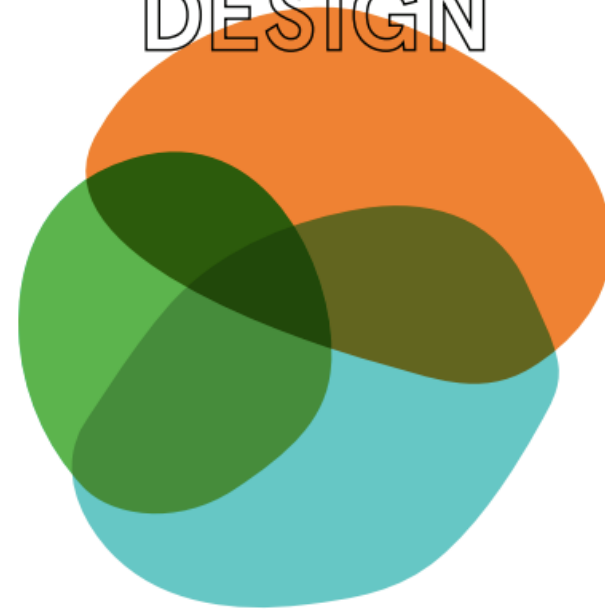


<https://www.ideo.org/>

Free guide



THE FIELD GUIDE TO HUMAN- CENTERED DESIGN



DESIGN KIT

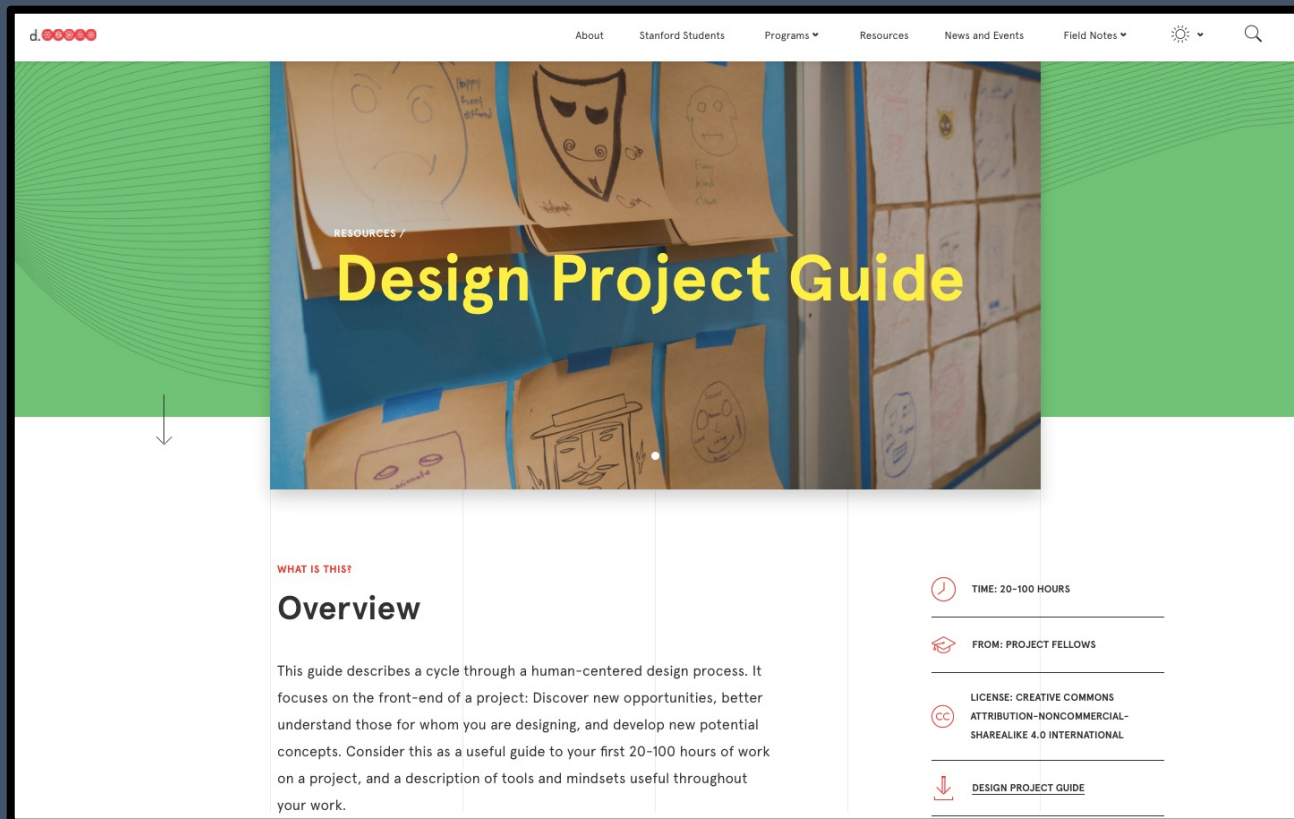
Welcome

EXPLORE THE STANFORD D.SCHOOL



<https://dschool.stanford.edu/>

Free guide



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DESIGN PROJECT GUIDE

This guide describes a cycle through a human-centered design process. It focuses on the front-end of a project: Discover new opportunities, better understand those for whom you are designing, and develop new potential concepts. Consider this as a useful guide to your first 20-100 hours of work on a project.

<https://stanford.io/3qyxOoL>

Prototyping tools

Prototyping is about design, not tools

Lecture: Prototyping is about design, not tools
Module: Prototyping tools

Prototyping is about design

- **Prototyping** is not **about** the **tool**.
 - ✓ It's **about** the **design**!
- The **tool supports** the **design** and **help** you get the **answers** you **need**
 - ✓ but the **design** is the **primary focus** !
- **Rationale:**
 - ✓ There is **no all-encompassing perfect tool** for **prototyping**.

Prototyping tools

- Find the **best tool** for your **workflow** and your **personal process**.
- You **don't have to know** every tool.
 - ✓ Find the **tools** and **process** that **work for** you and that **you** feel **comfortable** with, so you can **focus** on the **design**.

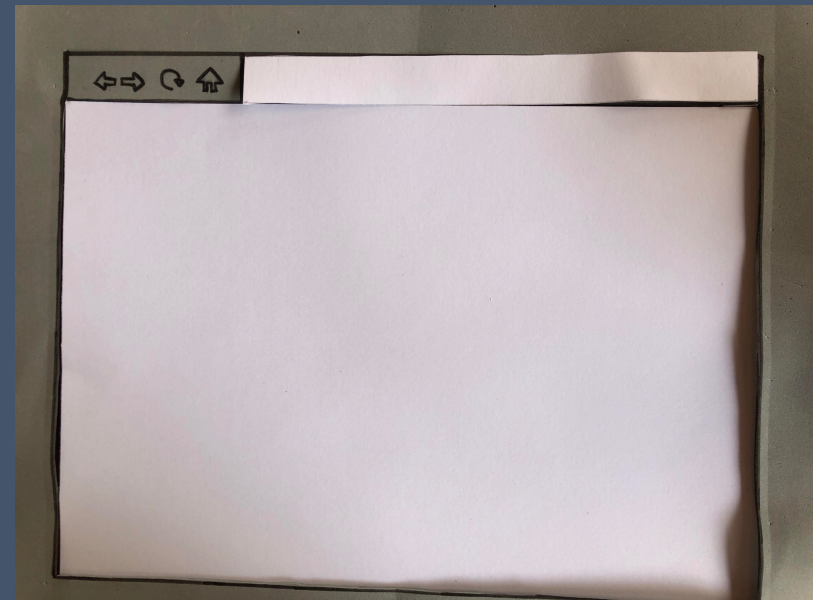
A paper prototype walkthrough

(Hostel Sparkling Water)

Lecture: A paper prototype walkthrough - Hostel Sparkling Water
Module: Prototyping tools

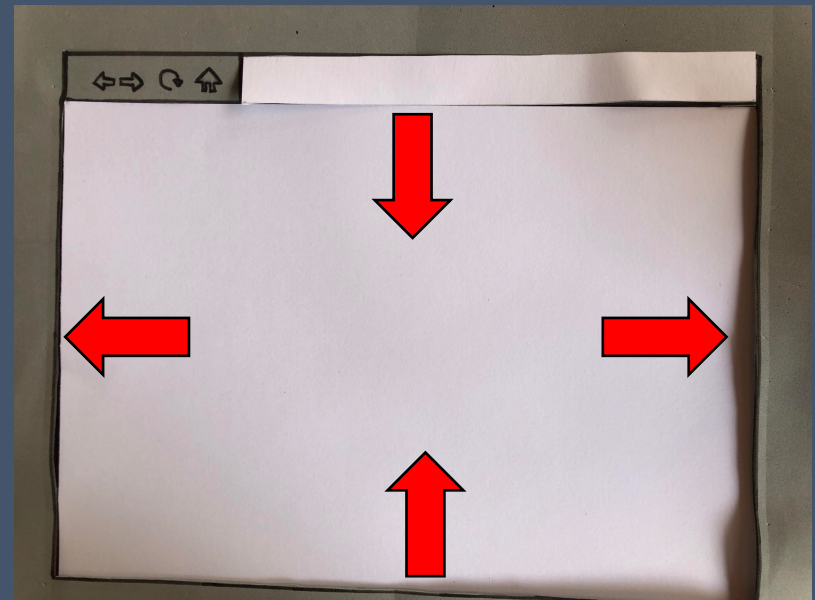
Paper prototypes

- In this example, we created a frame simulating a web browser and some blank A4 papers out of the view of the frame to draw our sketches.



Paper prototypes

- This design supports a scroll from left to right, right to left, top to bottom, and bottom-up.
 - ✓ By replacing the A4 paper during demonstration to the user.



Paper prototypes

- Your **sketches on paper** can be **detailed** or **high level**.
 - ✓ It really **depends** on how much **information** you want to **transmit** to the **users** and **what you want to get feedback on**.
- In the **example that follows**, we are **interested** in the **overall flow**.
 - ✓ We are **not going** to **get into too much detail** of the **screens**.

Example

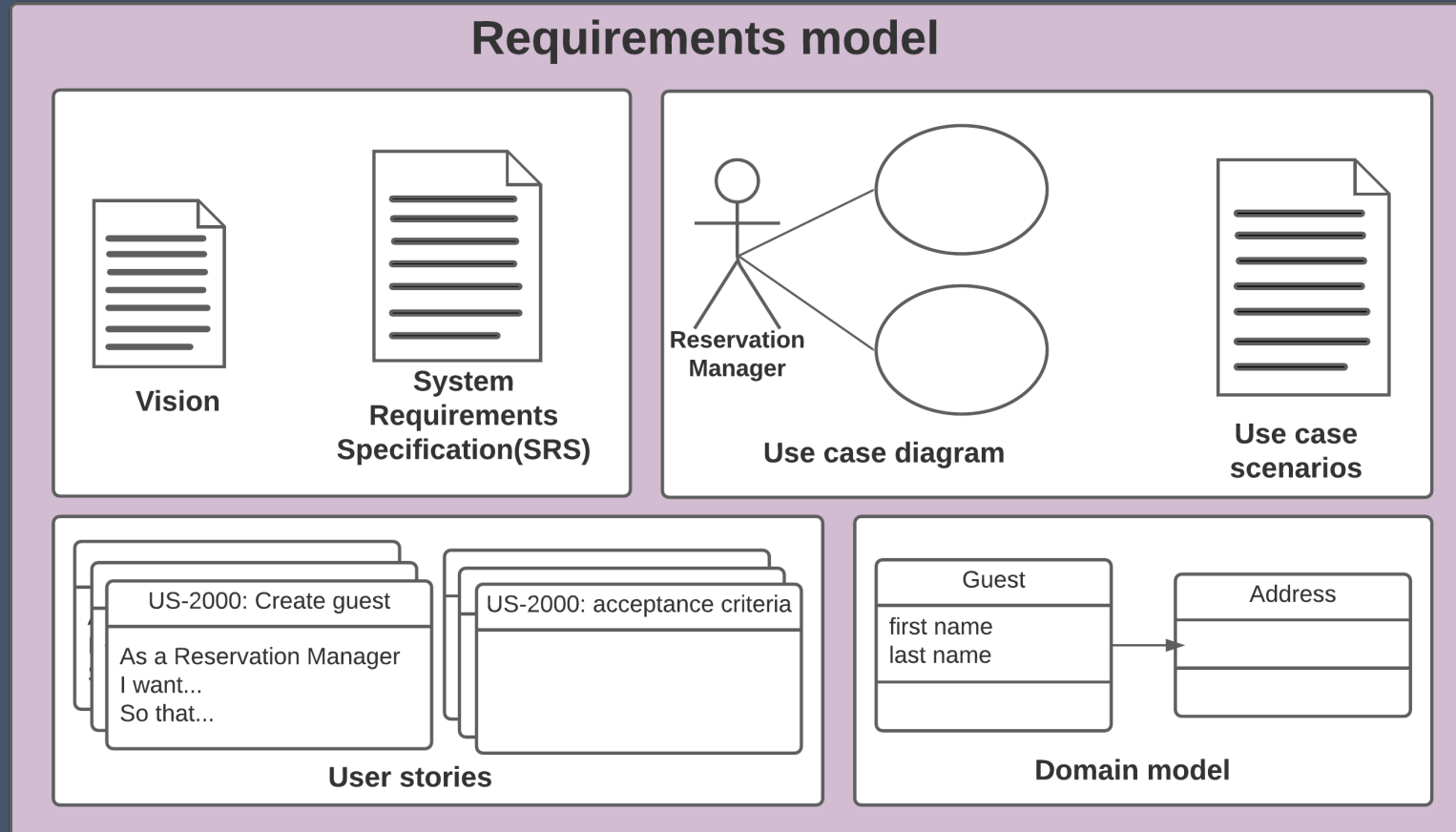
(Hostel Sparkling Water)

Example – Hostel Sparkling Water

- Let's **simulate** a **low-fidelity prototype** made of **sketches** as if we would be **putting** it in **front** of a **user**.
- **Suppose** we are **interested** in **how easy** is to **access** the **hostel services**:
 - ✓ Is it **easy** to **find** the **services** offered by the **hostel**?
 - ✓ Is it **easy** to **follow** the **flow** to **list** all **guests** in the **hostel**?
 - ✓ Is it **easy** to **create** a **new guest** in the **hostel**?

Requirements model & prototyping

- To set the context for this test, it's a good idea to give a scenario to the user.
- If you have some artifacts from a Requirements model in hands, you can use them to help during the walkthrough.



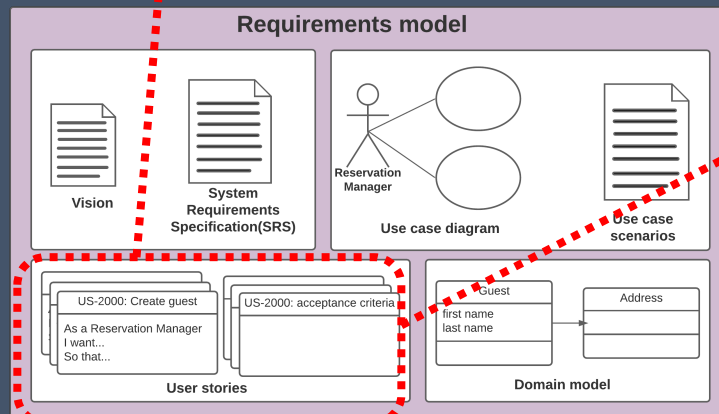
US-1055

**(View consolidated information about the hostel
and it's offered services)**

US-1055: View consolidated information about the hostel and it's offered services

As a **Visitor** to the **hostel Sparkling Water** website,
I want to see in a consolidated manner the **services offered** by the **hostel**
so that I can **select a service** I wish.

Let's build a sketch for this US



US-1055: Acceptance criteria

- The **consolidated information** must be in the **homepage** of the **hostel website**
- **Services offered** in the **homepage**:
 - **Guest** management, **Rooms** management, **Reservation** management
 - **Login/logout**
- **Information** available in the **homepage**:
 - **Brief history** of the **hostel Sparkling Water**.
 - Information about the **Park**, main attraction in **town**.
 - Information of **how** to **get** to the **hostel**.
 - **Contact information/address**.

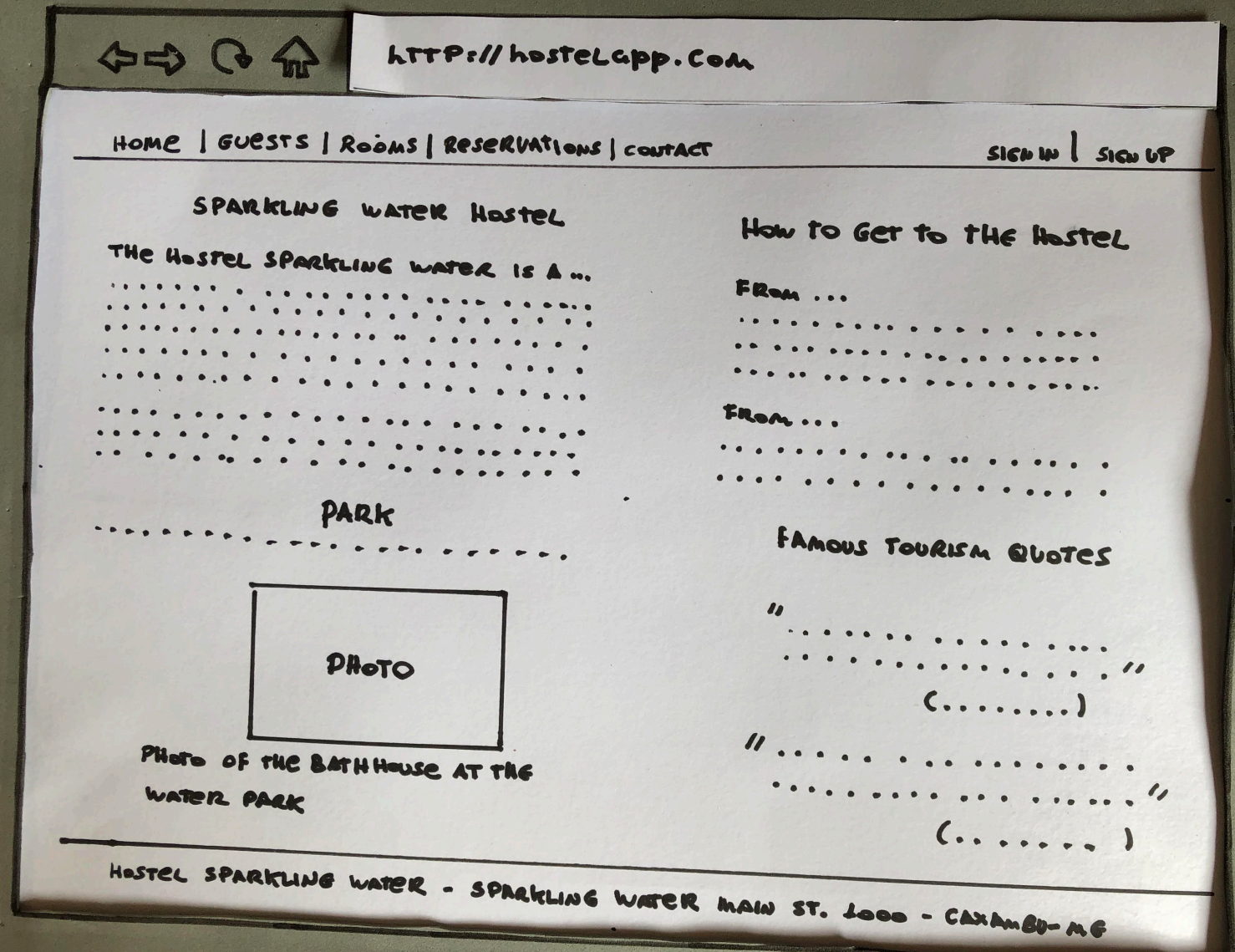
US-1055: View consolidated information about the hostel and it's offered services

As a Visitor to the hostel Sparkling Water website, I want to see in a consolidated manner the services offered by the hostel so that I can select a service I wish.

Hostel's homepage

US-1055: Acceptance criteria

- The consolidated information must be in the homepage of the hostel website
- Services offered in the homepage:
 - Guest management, Rooms management, Reservation management
 - Login/logout
- Information available in the homepage:
 - Brief history of the hostel Sparkling Water.
 - Information about the Park, main attraction in town.
 - Information of how to get to the hostel.
 - Contact information/address.



US-1648

(Log in)

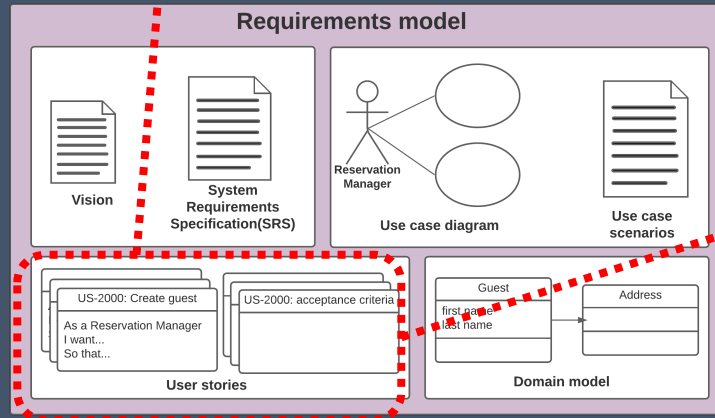
US-1648: Log in

As a Reservation manager

I want to login in the hostelApp

in order to create reservations for **new guests** and view a **list** with **all guests** that have already stayed at hostel.

In the same way,
let's build a sketch
for the US-1648



US-1648: Acceptance criteria

- A **login** has a **username** and a **password**.
- There must be an **explicit option** to remember the **Reservation manager** credentials.
- An **incorrect login message** must be shown for incorrect **username** and/or **password**.
- The **password** must be **at least 8 characters** long.

Sketch for US-1648

The sketch depicts a web browser window with a navigation bar at the top containing links: HOME | GUESTS | ROOMS | RESERVATIONS | CONTACT. The address bar shows the URL HTTP://HOSTELAPP.COM/LOGIN. The main content area is titled 'HOSTEL SPARKLING WATER - EMPLOYEE LOGIN' and contains two input fields labeled 'USERNAME' and 'PASSWORD', followed by a 'SIGN IN' button. The footer of the page reads 'HOSTEL - SPARKLING WATER MAIN ST. 1000 - CORANGUITE'.

US-1648: Acceptance criteria

- A login has a username and a password.
- There must be an explicit option to remember the Reservation manager credentials.
- An incorrect login message must be shown for incorrect username and/or password.
- The password must be at least 8 characters long.

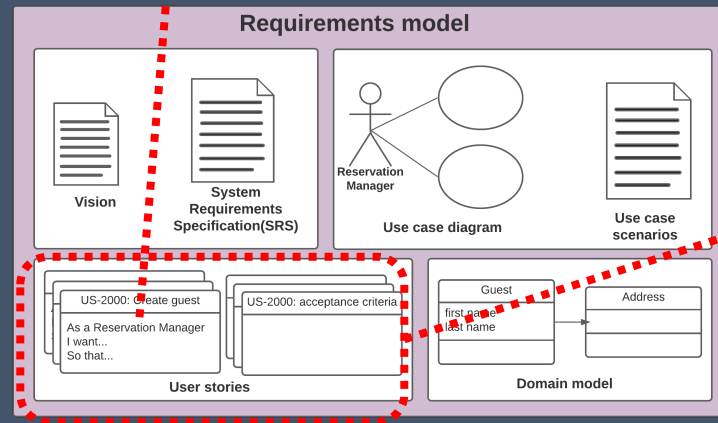
US-1112

(List guests)

US-1112: List guests

As a Reservation Manager,
I want to see a **list** with **all guests records** of the **Hostel**
in order to better know our **guests** and **communicate**
with them.

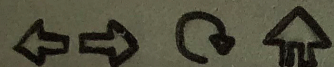
Now, for the the US-1112



US-1112: Acceptance criteria

- The **list** of **guests** may be **ordered** by the following guest data: **name**, **last name** and **date of birth**.

List of guests



http://www.HOTELAPP.com/GUESTS

Home | GUESTS | Rooms | Reservations | CONTACT

LOGOUT ~~SHOW~~ ~~SIGN UP~~

GUESTS
CUSTOMERS

ADD NEW GUESTS

SHOW 100 ENTRIES

SEARCH

#	PHONE NUMBER	FIRST NAME	LAST NAME	EMAIL
1	+1-205-555-0178	JAY	GATSBY	JAY@GMAIL.COM
2	+1-303-555-0137	HOLBEN	CAULFIELD	holben@MIT.EDU
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US-1105

(Create a guest)

US-1105: Create a guest

As a Reservation Manager

I want to create a guest

in order to keep a history of all guests who have been in the Sparkling Water Hostel since its inauguration.

User story referencing a use case scenario

US-1105: Acceptance criteria

- **See scenario in the use case model.**
- A **guest** has the following fields: **first name**, **last name**, **email**, **phone number**, **address**, **country**, **state**.
- The **name**, **last name**, **address**, **state**, **country**, and **email** fields are **required** and **cannot be empty**.
- The name must be **at least 2 characters**
- The **Reservation manager** must be notified when a literal in a field violates the rules (**required**, **number of chars**, **mandatory fields**, etc)
- The system must verify **Invalid date format** (the format for dates used in the system is YYYY/MM/DD)
- The **same guest cannot** be created **twice**. A message "**guest already exists**" must be shown in this case. A guest can be uniquely identified by its **name + last name + e-mail**.
- ...
- ...

Scenario: main flow for creating a guest

Use case scenario: Create a Guest

Pre-condition: a screen with an option “Create new Guest” is visible to the Reservation Manager

Main flow:

- 1.The Reservation Manager selects the option Create new Guest
- 2.The system show a Form requesting Guest data
- 3.The Reservation Manager enters the form data
- 4.The systems shows a message Guest created sucessfully.

Alternative flow:

....

....

Larman's System Sequence Diagram

Use case scenario: Create a Guest

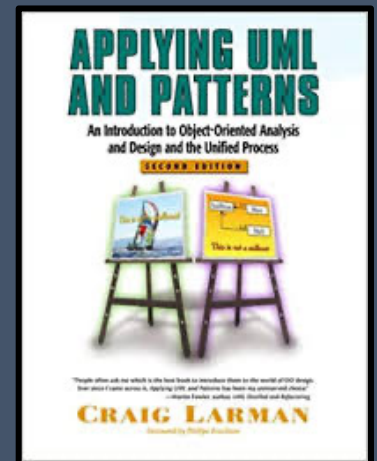
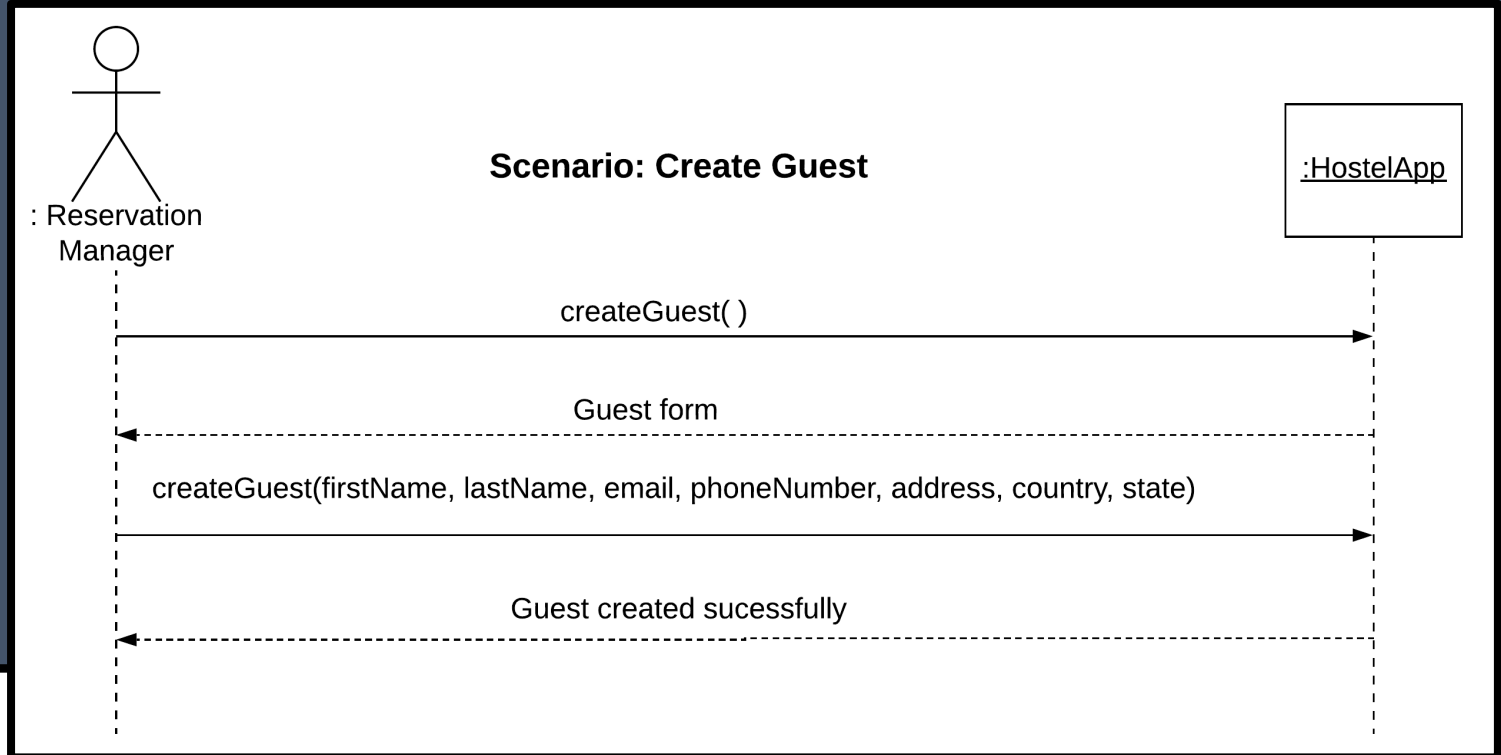
Pre-condition: a screen with an option "Create new Guest" is visible to the Reservation Manager

Main flow:

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Alternative flow:

....
....



Sketch for US-1105 (1/4)

← → ↺ 🏠 <http://www.HOSTELAPP.com/GUESTS>

Home | GUESTS | Rooms | RESERVATIONS | CONTACT LOGOUT ~~SHOW IN~~ ~~SIGN UP~~

GUESTS CUSTOMERS ADD NEW GUESTS

SHOW 10 ENTRIES SEARCH

#	PHONE NUMBER	FIRST NAME	LAST NAME	EMAIL
1	+1-205-555-0178	JAY	GATSBY	JAY@GMAIL.COM
2	+1-303-555-0137	HOLBEN	CAULFIELD	HOLBEN@MIT.EDU
3		..	..	..
4		..	..	..
5		..	..	..
6		..	..	..
7		..	..	..
8		..	..	..
9		..	..	..
10	+1-860-555-0154	LI	..	..

SHOWING 1 to 10 of 500 entries

HOSTEL SPARKLING WATER - SPARKLING

Use case scenario: Create a Guest

Pre-condition: a screen with an option "Create new Guest" is visible to the Reservation Manager

Main flow:

1. The Reservation Manager selects the option Create new Guest
2. The system show a Form requesting Guest data
3. The Reservation Manager enters the form data
4. The systems shows a message Guest created sucessfully.

Alternative flow:

....

....

Sketch for US-1105 (1/4)

← → ↺ 🏠 <http://www.HOSTELAPP.com/GUESTS>

Home | GUESTS | Rooms | RESERVATIONS | CONTACT LOGOUT ~~SHOW IN~~ ~~SIGN UP~~

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4		..	..	..
5		..	..	..
6		..	..	..
7		..	..	..
8		..	..	..
9		..	..	..
10	+1-860-555-0154	LI		

SHOWING 1 to 10 of 500 entries

HOSTEL SPARKLING WATER - SPARKLING

Use case scenario: Create a Guest

Pre-condition: a screen with an option "Create new Guest" is visible to the Reservation Manager

Main flow:

1. The Reservation Manager selects the option Create new Guest
2. The system show a Form requesting Guest data
3. The Reservation Manager enters the form data
4. The systems shows a message Guest created sucessfully.

Alternative flow:

....

....

Sketch for US-1105 (2/4)

HTTP://WWW.HOSTELAPP.COM/GUESTS/ADD

Home | GUESTS | ROOMS | RESERVATIONS | CONTACT

Logout

GUEST

FIRST NAME

LAST NAME

E-MAIL ADDRESS

PHONE NUMBER

ADDRESS

COUNTRY

STATE

SUBMIT

RETURN TO ALL GUESTS

HOTEL SPARKLING WATER - SPARKLING

Use case scenario: Create a Guest

Pre-condition:

a screen with an option "Create new Guest" is visible
the Reservation Manager

Main flow:

1. The Reservation Manager selects the option Create new Guest

2. The system show a Form requesting Guest data

3. The Reservation Manager enters the form data

4. The systems shows a message Guest created sucessfully.

Alternative flow:

Sketch for US-1105 (3/4)

← → ↺ ↻

HTTP://WWW.HOTELAPP.COM/GUESTS/ADD

HOME | GUESTS | ROOMS | RESERVATIONS | CONTACT

LOGOUT

GUEST

REGION

RASKOLNIKOV

RASKOLNIKOV@GMAIL.COM

+7-850-333-2272

SAINT PETERSBURG STREET, 1523

RUSSIA

RA

SUBMIT

RETURN TO ALL GUESTS

HOTEL SPARKLING WATER - SPARKLING

Use case scenario: Create a Guest

Pre-condition: a screen with an option "Create new Guest" is visible to the Reservation Manager

Main flow:

1. The Reservation Manager selects the option Create new Guest

2. The system show a Form requesting Guest data

3. The Reservation Manager enters the form data

4. The systems shows a message Guest created sucessfully.

Alternative flow:

....

....

Sketch for US-1105 (4/4)

← → ↺ ↻

HTTP://WWW.HOSTELAPP.COM/GUESTS/ADD

HOME | GUESTS | ROOMS | RESERVATIONS | CONTACT

LOGOUT

GUEST

RODION

RASKOLNIKOV

RASKOLNIKOV@GMAIL.COM

SAINT PETERSBURG ST

RUSSIA

SUBMIT

RETURN TO ALL GUESTS

HOSTEL SPARKLING WATER - SPARKLING

GUEST CREATED SUCCESSFULLY!

Use case scenario: Create a Guest

Pre-condition:

a screen with an option "Create new Guest" is visible to the Reservation Manager

Main flow:

1.The Reservation Manager selects the option Create new Guest

2.The system show a Form requesting Guest data

3.The Reservation Manager enters the form data

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Alternative flow:

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Low fidelity prototypes for the HostelApp using LucidChart

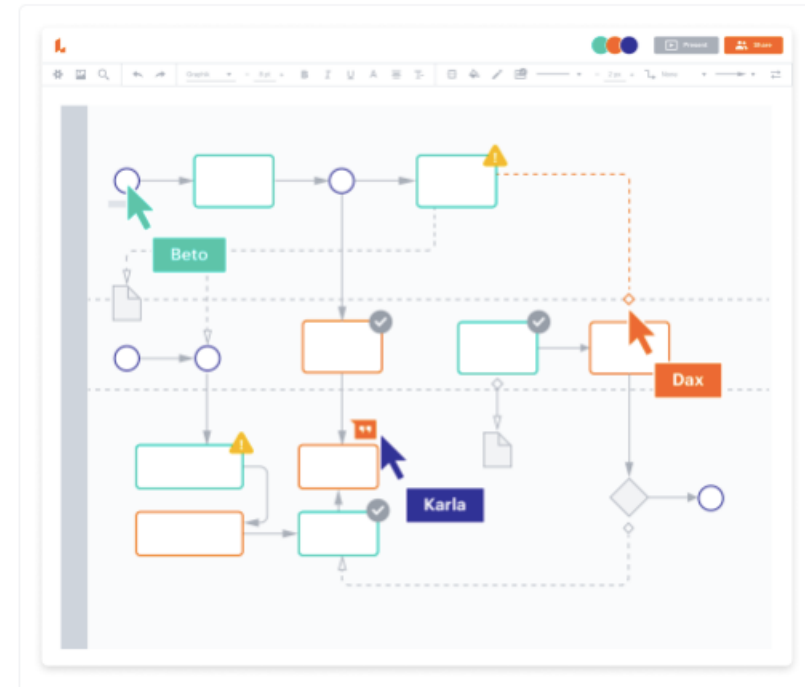
Lecture: Low fidelity prototypes for the HostelApp using LucidChart

Module: Prototyping tools

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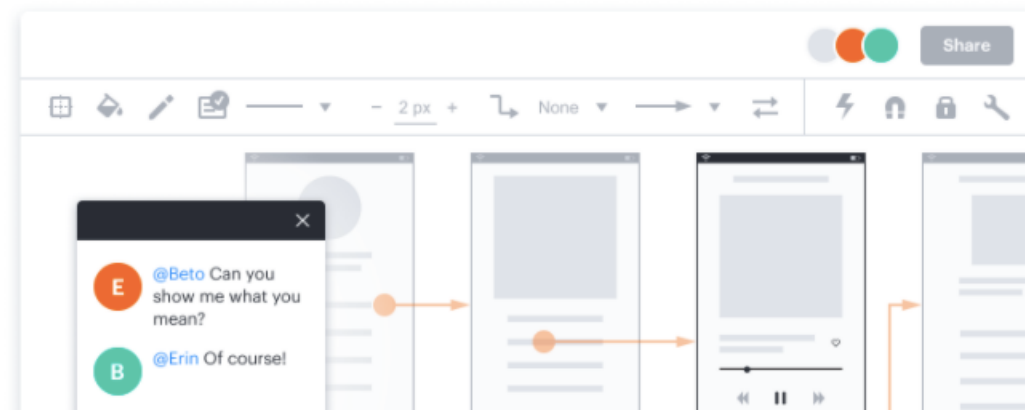
Sign up free



Diagramming and flowcharting

Software and systems design

Org chart design



<http://lucidchart.com>

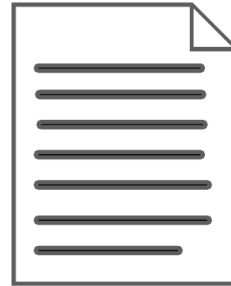
**We used LucidChart to create some
low-fidelity prototypes for the
Hostel Sparkling Water**

Requirements model

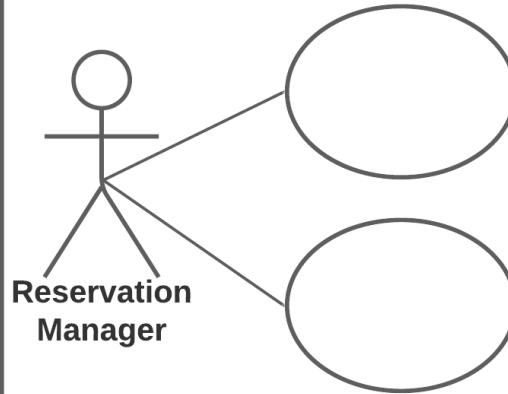
Requirements model



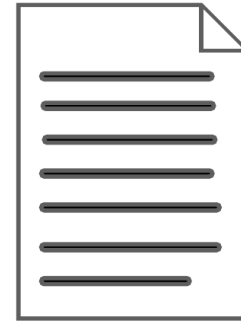
Vision



System
Requirements
Specification(SRS)



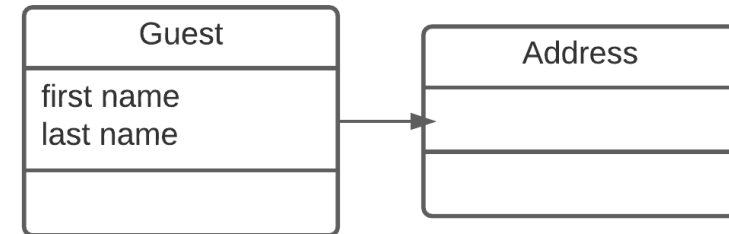
Use case diagram



Use case
scenarios



User stories



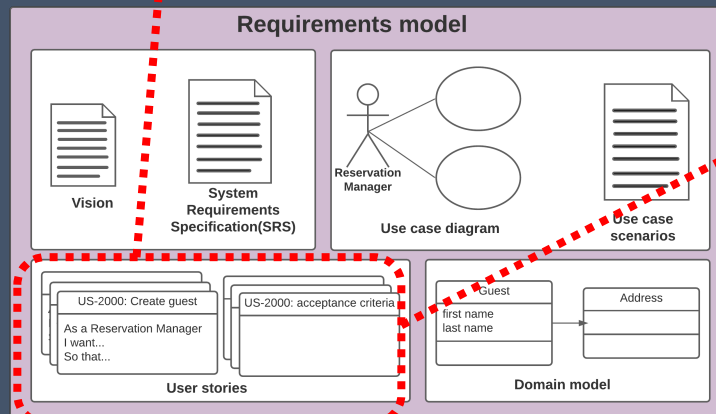
Domain model

US-1055

**(View consolidated information about the hostel
and it's offered services)**

US-1055: View consolidated information about the hostel and it's offered services

As a Visitor to the **hostel Sparkling Water** website,
I want to see in a consolidated manner the **services offered** by the **hostel**
so that I can **select a service** I wish.



US-1055: Acceptance criteria

- The **consolidated information** must be in the **homepage** of the **hostel website**
- **Services offered** in the **homepage**:
 - **Guest** management, **Rooms** management, **Reservation** management
 - **Login/logout**
- **Information** available in the **homepage**:
 - **Brief history** of the **hostel Sparkling Water**.
 - Information about the **Park**, main attraction in **town**.
 - Information of **how** to **get** to the **hostel**.
 - **Contact information/address**.

Home page US-1055

US-1055: Acceptance criteria

- The **consolidated information** must be in the **homepage** of the **hostel website**
- **Services offered** in the **homepage**:
 - **Guest** management, **Rooms** management, **Reservation** management
 - **Login/logout**
- **Information** available in the **homepage**:
 - **Brief history** of the **hostel Sparkling Water**.
 - Information about the **Park**, main attraction in **town**.
 - Information of **how to get to the hostel**.
 - **Contact information/address**.

Sparkling Water Hostel Application

← → ↺ ⬆

http://hostelapp.com

Home | Customers | Rooms | Reservations | Contact

Sign in | Sign up

Sparkling Water Hostel

The *hostel Sparkling Water* is a fictitious hostel located in Caxambu-MG, Brazil. Caxambu is a small town (its population in 2020 was estimated at 22,000), renowned for its spa which has twelve sources of sparking mineral water flowing 24 hours a day and a cold-water geyser. Caxambu is the only place in the world with 12 mineral water springs, each with different chemical properties.

How to get to the Hostel?

From Rio de Janeiro
Via BR-116 (Presidente Dutra) west past Resende-RJ to the village of Engenheiro Passos-RJ and take BR-354 north.

From Belo Horizonte or São Paulo
Via BR-381 to Campanha-MG, then west on BR-267.

Park

The Water Park is the main attraction in town.



Photo of the bathhouse located at the Water Park

Famous tourism quotes

"Never go on trips with anyone you do not love." *Ernest Hemingway*

"To travel is to discover that everyone is wrong about other countries." *Aldous Huxley*

Sparkling Water Hostel . Sparkling water main st. 1000. Caxambu-MG.

US-1648

(Log in)

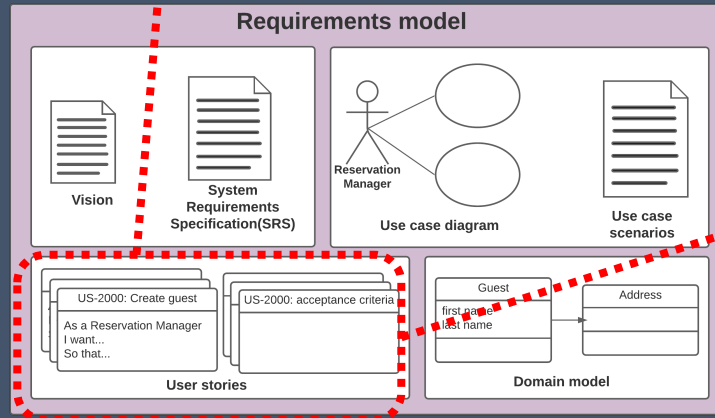
US-1648: Log in

As a Reservation manager

I want to login in the hostelApp

in order to create reservations for **new guests** and view a list with all guests that have already stayed at hostel

US-1648: Log in



US-1648: Acceptance criteria

- A **login** has a **username** and a **password**.
- there must be an **explicit option** to remember the **Reservation manager** credentials.
- An **incorrect login message** must be shown for incorrect **username** and/or **password**.
- The **password** must be **at least 8 characters** long.

Login page for Employees US-1648

Sparkling Water Hostel Application

← → ↻ 🏠

Home Customers Rooms Reservations Contact [Login](#)

Hostel Sparkling Water - Employee Login

Username

Password

[Sign in](#)

US-1648: Acceptance criteria

- A login has a username and a password.
- There must be an explicit option to remember the Reservation manager credentials.
- An incorrect login message must be shown for incorrect username and/or password.
- The password must be at least 8 characters long.

Sparkling Water Hostel .

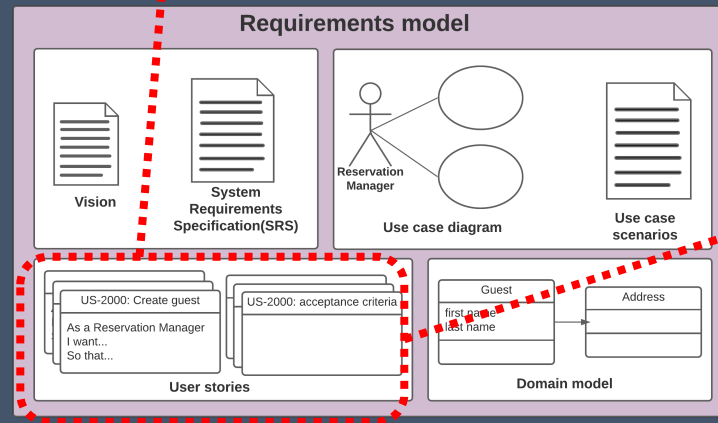
US-1112

(List guests)

US-1112: List guests

As a **Reservation Manager**,
I want to see a **list** with **all guests records** of the **Hostel**
in order to better know our **guests** and **communicate**
with them.

Now, for the the US-1112



US-1112: Acceptance criteria

- The **list** of **guests** may be **ordered** by the following guest data: **name**, **last name** and **date of birth**.

List of guests

Sparkling Water Hostel Application					
← → ↺ ⬆ http://hostelapp.com/guests					
HomeCustomersRoomsReservationsContact					Logout
Guests Add new guest					
Show10entries Search:					
#	Phone Number	First	Last	Email Address	
11	+1-847-555-0127	Holly	Golightly	golightly@gmail.com	
12	+1-808-555-0162	Gregor	Samsa	samsa@yahoo.com	
13	+1-847-555-0151	Aureliano	Buendia	buendia@yahoo.com	
14	+1-785-555-0189	Clarissa	Dalloway	dalloway@gmail.com	
15	+1-920-555-0109	Ignatius	Reilly	reilly@gmail.com	
16	+1-785-555-0132	George	Smiley	smiley@gmail.com	
17	+1-860-555-0146	Winnie	Pooh	pooh@yahoo.com	
18	+1-510-555-0112	Bigger	Thomas	thomas@hotmail.com	
19	+1-605-555-0121	Nick	Adams	adams@gmail.com	
20	+1-510-555-0187	Scarlett	OHara	ohara@gmail.com	
23 entries					Previous123Next
Sparkling water main st. 1000. Caxambu-MG.					

US-1112: List guests

As a Reservation Manager,
I want to see a list with all guests records of the Hostel
in order to better know our guests and communicate
with them.

US-1105

(Create a guest)

US-1105: Create a guest

As a Reservation Manager

I want to create a guest

in order to keep a history of all guests who have been in the Sparkling Water Hostel since its inauguration.

User story referencing a use case scenario

US-1105: Acceptance criteria

- **See scenario in the use case model.**
- A **guest** has the following fields: **first name**, **last name**, **email**, **phone number**, **address**, **country**, **state**.
- The **name**, **last name**, **address**, **state**, **country**, and **email** fields are **required** and **cannot be empty**.
- The name must be **at least 2 characters**
- The **Reservation manager** must be notified when a literal in a field violates the rules (**required**, **number of chars**, **mandatory fields**, etc)
- The system must verify **Invalid date format** (the format for dates used in the system is YYYY/MM/DD)
- The **same guest cannot** be created **twice**. A message "**guest already exists**" must be shown in this case. A guest can be uniquely identified by its **name + last name + e-mail**.
- ...
- ...

Create guests

Sparkling Water Hostel Application

← → ↻ ⬆

http://hostelapp.com/customers/add

HomeGuestsRoomsReservationsContactLogin

Create guest

First Name

Last Name

Email Address

Phone Number

Address

Country

State

Save

Use case scenario: Create a Guest

Pre-condition: a screen with an option “Create new Guest” is visible to the Reservation Manager

Main flow:

- 1.The Reservation Manager selects the option Create new Guest
- 2.The system show a Form requesting Guest data
- 3.The Reservation Manager enters the form data
- 4.The systems shows a message Guest created sucessfully.

Alternative flow:

....

....

Sparkling Water Hostel . Sparkling water main st. 1000. Caxambu-MG.

Other software tools

Lecture: Other software tools

Module: Prototyping tools

Prototyping tools

- POP – Prototyping On Paper
- Sketch
- Moqups
- Balsamiq
- Invision
- Axure





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Prototyping on paper

POP helps you transform your pen and paper ideas into an interactive iPhone or Android prototype.



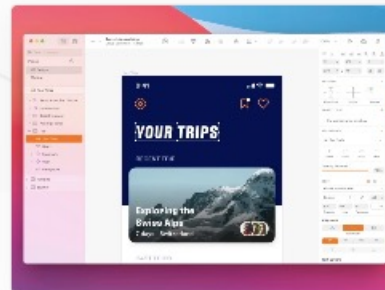
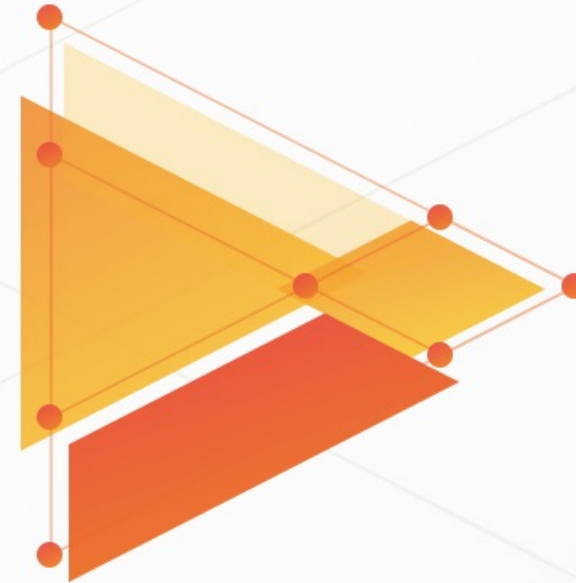
<https://marvelapp.com/pop>

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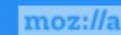
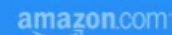
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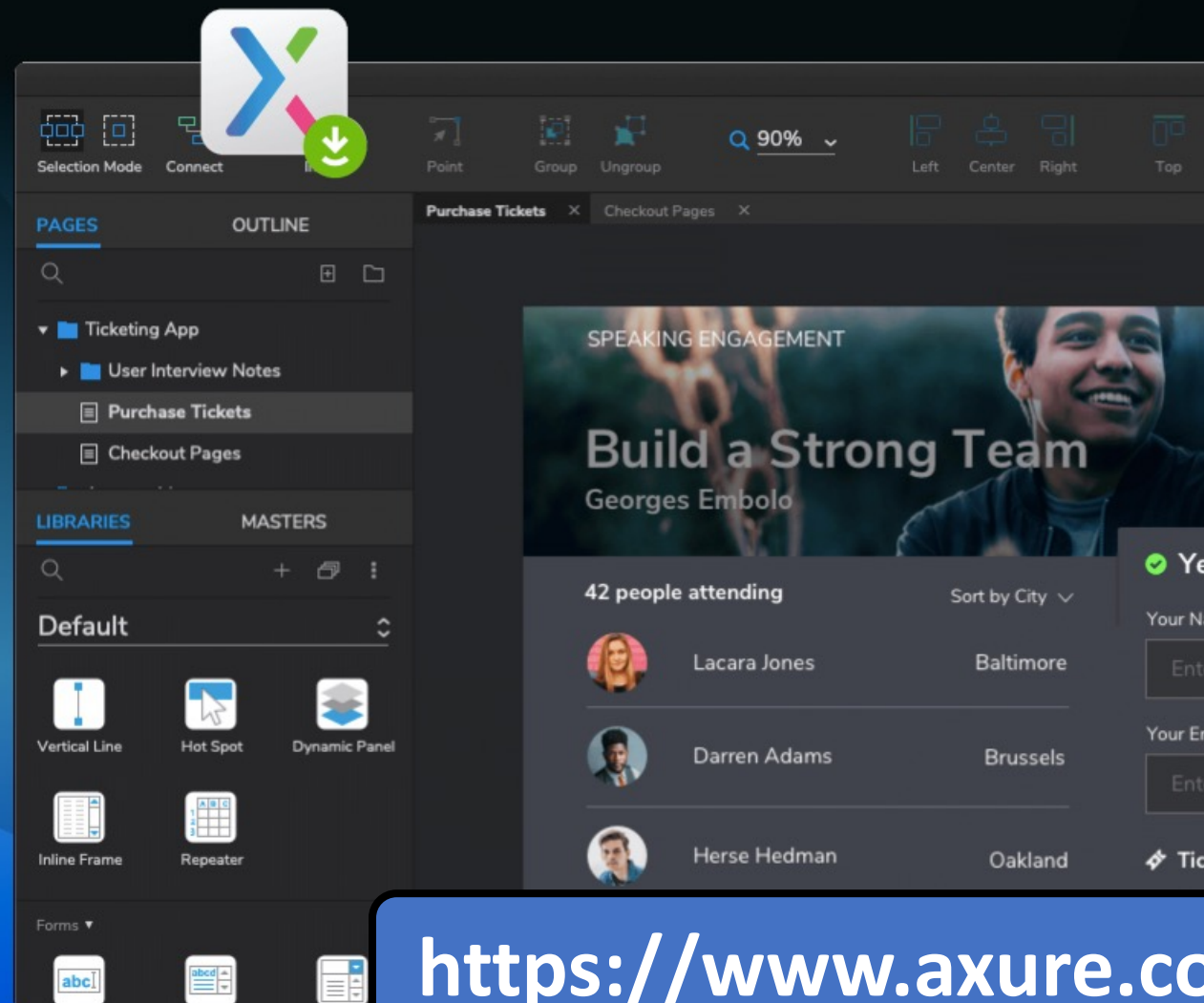


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